

Lab06 Homework

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Improving analysis code by writing functions

Q6a. How would you generalize the original code above to work with any set of input protein structures?

This code creates a data frame and attempts to normalize each column and repeating the same normalization formula multiple times for each column.

```
# A. Can you improve this analysis code?
df <- data.frame(a=1:10, b=seq(200,400,length=10),c=11:20,d=NA)
df$a <- (df$a - min(df$a)) / (max(df$a) - min(df$a))
df$b <- (df$b - min(df$a)) / (max(df$b) - min(df$b))
df$c <- (df$c - min(df$c)) / (max(df$c) - min(df$c))
df$d <- (df$d - min(df$d)) / (max(df$a) - min(df$d))
```

One way that better improve this code is to create a function that avoids the repeat.

```
normalize <- function(x) {
  (x - min(x)) / (max(x) - min(x))
}

dfnorm <- data.frame(a=1:10, b=seq(200,400,length=10), c=11:20, d=NA)

dfnorm$a <- normalize(dfnorm$a)
dfnorm$b <- normalize(dfnorm$b)
dfnorm$c <- normalize(dfnorm$c)
dfnorm$d <- normalize(dfnorm$d)
```